

Reverse Engineering and Panic Pricing in Under-Over Markets

1. Epistemological Foundation of the Poisson Paradox

The Poisson Paradox framework is a mathematical infrastructure designed to extract Expected Value (+EV) by exploiting the structural rigidities and genetic flaws inherent in institutional Dixon-Coles pricing models. While traditional sports betting relies on "predictive heuristics" to forecast match outcomes, this model utilizes "stochastic misalignment exploitation." It specifically targets the algorithmic inability of market makers to modulate risk during "stochastic shocks"—early goals that trigger a total collapse of the Tau (τ) parameter and a subsequent irrazionale overshoot in pricing.

Key Principle: The Poisson Paradox model targets algorithmic collapse rather than match outcomes. It capitalizes on the γ factor (Panic Pricing) where the market maker's over-correction creates a mathematical delta between offered odds and fair probability.

2. The Hard Trigger Gate: Pre-Match Market Compression

Operational entry is dictated by the "Hard Trigger Gate," occurring when pre-match Draw (X) odds collapse to ≤ 3.00 (with a 3.10 structural tolerance). This compression signals a "saturation of central probability" within the Dixon-Coles algorithm, forcing a non-linear redistribution of probability toward the distribution tails. This results in the hyper-protection of the Under 2.5 market.

Pre-Match Trigger Coordinates	Market Metric	Target Coordinate	Algorithm Reaction	Target CLV
:--- :--- :--- :---	Draw Odds (X) ≤ 3.00 (3.10 tol)	Saturation of central probability	N/A	
Under 2.5 Quota 1.44 – 1.53	Non-linear probability redistribution	1.50 - 1.65		Over 2.5 Quota ≥ 2.75 – 3.00
Asymmetric "bulge" formation	N/A		Asian Under 2.0 ~ 2.15	Target Anchor for efficient markets 2.15

3. Structural Branching of Path B (Strada B)

Path B is not a linear vector but a bifurcated node that adapts to the "atmospheric configurations" of the match. Depending on the manifestation of live variance, the system engages one of two distinct operational states:

1. **Strada B (Null Variance):** Activated if the score remains 0-0 at 30 minutes. This branch exploits **Theta (Time Decay)** by acquiring Over 2.5 positions as the initial price "bulge" expands due to the exit of "dumb money" and lack of market shocks.
2. **Tranche Alpha (Active Variance):** Activated by an early goal within the 0-25 minute window. This branch utilizes **Inversion Logic**, pivoting from the Over market to exploit **Panic Pricing** in the Under 2.5 and Under 3.5 lines. It targets the "aberrant" prices created by algorithmic over-correction following a stochastic shock.

4. Mathematical Architecture of Reverse Engineering (RE)

The RE algorithm bypasses the limitations of high-frequency tick-by-tick data by analytically deriving the bookmaker's internal risk parameters through three phases:

1. **\lambda Root Extraction:** Using the pre-match Under 2.5 Closing Line Value (CLV) to solve the Poisson cumulative distribution function for total expected goals (λ_{pre}). A CLV of 1.62 implies $\lambda_{pre} = 2.30$.
2. **Time Decay & TAC:** Adjusting the expectation for the remaining match duration (t_{rem}) and applying a **Tactical Aggression Coefficient (TAC)**:
 - o $\lambda_t = \lambda_{pre} * (t_{rem} / 90)$
 - o $\lambda_{rem} = \lambda_t * 1.05$
3. **Fair Odds Derivation:** Calculating the theoretical "true" price of the Under markets post-shock, which serves as the baseline to identify market overshoot.

The 1.05 Tactical Aggression Coefficient acts as a 5% inertia factor to account for momentum and tactical frenzy immediately following a goal.

5. Phenomenology of Panic Pricing (γ) and Temporal Zones

Panic Pricing is driven by the multiplier γ , defined by the ratio of algorithmic rigidity (k) to time (t). This creates an "Overshoot" where the bookmaker price exceeds Fair Odds.

Structural Tilt ($t \rightarrow 1$)

In the event of a goal in the opening minute, the denominator t approaches zero, pushing the panic function toward its vertical asymptote. This results in a total collapse of the τ (Tau) parameter as the bookmaker's server enters a state of algorithmic tilt, causing Under 3.5 quotas to explode.

The Strike Zone (6' – 15')

The "Golden Zone" for **Tranche Alpha**. While time decay is minimal, the k/t ratio is sufficient to force a "Rimbalzo Anomalo" (Anomalous Bounce). Bookmakers hyper-protect against a blowout, pushing Under 2.5 odds to irrational levels (e.g., from 1.60 to 3.50).

Transition Phase (15' – 20')

The market begins absorbing the linear decay of λ_t . The k/t multiplier dampens, and the overshoot becomes less pronounced as the "panic" normalizes toward the Fair Odds baseline.

Safety Threshold (> 20' - 35')

The k/t multiplier effectively deactivates. The "bug" is no longer exploitable as the market returns to stochastic linearity. The system disengages **Tranche Alpha** and demands a phase shift to the Rent Extraction of Path C.

6. RE Simulation Table: Market Dynamics (2025-2026 Dataset)

The following data reflects the simulation of the 2025-2026 dataset, demonstrating the relationship between the minute of the goal (t) and the extracted Edge.

Match Event	CLV (U2.5)	Goal (t)	Fair Odds	Under 2.5 Spike	Spike Propagato U3.5	Edge
Lecce-Cagliari	1.53	8'	~2.15	2.85 - 2.95	1.65 - 1.75	+30.0%
Genoa-Lazio	1.62	11'	~2.55	3.30 - 3.55	1.95 - 2.05	+34.3%
Verona-Cagliari	1.53	14'	~2.09	2.65 - 2.80	1.50 - 1.60	+29.1%

Torino-Cremonese	1.67	26'	~2.10	2.25 - 2.30	1.30 - 1.35	< 5.0%
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7. Analysis of the Joint Movement: Structural Tilt vs. Safety Threshold

The effectiveness of Reverse Engineering is strictly time-dependent.

- **Case Study: Genoa-Lazio (11')**: At $t=11$, the k/t ratio generates a massive overshoot. While Fair Odds were 2.55, the market offered 3.50. The system exploited the Dixon-Coles inability to modulate the bivariate dependence (τ) in real-time, yielding a **+34.3% Edge**.
- **Case Study: Torino-Cremonese (26')**: At $t=26$, the denominator t has grown sufficiently. The market spike stops near the Fair Odds (2.30 vs 2.10). Because the edge is $< 5\%$, the system recognizes that "Theta" has taken over and transitions to Path C.

8. Risk Management and the "Constraint Flip"

To navigate the "Overdispersion" typical of football markets, the model employs a strict capital allocation strategy:

- **Fractional Kelly Criterion:**
 - **1/4 Kelly:** Applied to speculative branches (**Tranche Alpha** and **Strada B**).
 - **1/8 Kelly:** Applied to Path C (Rent Extraction) to ensure capital preservation.
- **Constraint Flip:** A logical gate that automatically freezes speculative Path B operations during drawdowns. If the Monte Carlo convergence detects a drawdown approaching **17.89%**, the system prioritizes the high-probability yield of Path C.
- **Historical Performance:**
 - **Success Rate (Path C):** 91%
 - **Max Drawdown:** 17.89%
 - **Cumulative ROI:** +115%

9. Technical Conclusion: The "Soft Booky" Bug

Reverse Engineering is the necessary tool for bypassing the lack of tick-by-tick data. By solving for λ and isolating the panic multiplier γ , the model capitalizes on the "Soft Booky" bug. The failure of institutional models lies in their inability to modulate the bivariate dependence (τ) in real-time without breaking the "overround of guarantee." This framework converts that rigidity into a differential financial advantage.

Analyst Note: By anticipating the irrational overshoot in Under markets during the Strike Zone, the framework secures +EV positions that are mathematically insulated from match-specific noise and team-specific variance.